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Artificial Intelligence (AI) and the University

How AI Will Change Our Approach to Education

The second half of the 20th century and the beginning of the new century witnessed a dynamic expansion of education, especially in developing countries. There is no doubt that hundreds of millions of people worldwide have benefited from universal education in the last 80 years, since the end of World War II. The improvement of education, even in the poorest and most underdeveloped regions, is unquestionably one of the successes of global educational development. However, the majority of pupils and students worldwide choose the study of humanities, which involves the preparation and writing of a final paper: bachelor's, master's or doctoral thesis. As sociological research has revealed, the proportion of students demonstrating aptitude in the sciences, particularly in disciplines such as mathematics, physics, and chemistry, remains relatively constant in the population, which results in humanities students outnumbering science students.¹

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¹ Cf. Gunnar Heinsohn, *Wettkampf um die Klugen Kompetenz. Bildung und die Wohlfahrt der Nationen* (Zürich: Orell Füssli Verlag, 2019), 88.

The Creative Power of Artificial Intelligence

In light of the rapid progress in the field of artificial intelligence (AI), questions concerning the future direction of education, especially at the higher level, must be posed. The dynamic development of AI has undeniably given rise to situations in which it becomes challenging to determine whether a work of art is the result of human effort, or a creation generated by computer algorithms. The pervasive influence of “thinking machines” on our daily lives is already apparent and it is imperative to acknowledge the impact of AI on educational institutions, including schools and universities.² Toby Walsh, an Australian scientist with extensive experience in developing and researching new technologies, is hardly exaggerating when he offers his perspective on the issue:

These machines are set to alter our lives just as dramatically as the steam engine did at the start of the industrial age. They will change how we work, how we play, how we educate our children, how we treat the sick and elderly, and ultimately how humankind is remembered. They are likely our most transformative creations. [...] [It] will change how we relate to technology. It will change how we relate to each other. I would argue that it will even change how we perceive what it means to be human.³

² Cf. John McCarthy, “What is Artificial Intelligence?”? <https://www-formal.stanford.edu/jmc/whatisai.pdf> (accessed: March 2025). The term “thinking machines” is, of course, a metaphor, although it has unfortunately been popularized by the very expression “artificial intelligence” coined by the American scientist John McCarthy (1927–2011) who, together with other researchers, worked on algorithms in the 1950s. It is noteworthy that McCarthy established the Artificial Intelligence Laboratory at Stanford University, which is still in operation today as a museum dedicated to “thinking machines” and offers a comprehensive perspective on the evolution of this technology. See: <https://ai.stanford.edu/> (accessed: March 2025).

³ Toby Walsh, *It's Alive! Artificial Intelligence from the Logic Piano to Killer Robots* (Melbourne: La Trobe University Press, 2017), 9–11.

Observers of these technological changes are justified in posing the following question: will the advancement of artificial intelligence and its propensity to take over various human competencies inevitably result in the devaluation of work that is based on reading, assimilating knowledge, reasoning, discovery, writing and creation of scientific and literary works, as well as cultural artefacts? Does the development of AI bring us closer to the controversial, albeit probable, hypothesis that at a certain point the will and agency of human creators will become superfluous and even redundant? Will we reach a point where, as Jacek Dukaj suggests, the source of literature or art itself will no longer matter to us? The identity of the creator, whether human or machine, will become irrelevant. The sole criterion for evaluating the work will be its perception and reception by the beholder.⁴ The fact that we are already able to appreciate musical compositions reminiscent of Mozart, admire paintings similar to those in John Singer Sargent's collection, marvel at poems that bear resemblance to those of Eliot, and even read books that can compete with Agatha Christie's detective stories, indicates a shift in the artistic landscape, where the art of *pure logos*, once exclusively attributed to human ingenuity, is now subject to a redefined paradigm. The field of artificial intelligence no longer necessitates the inclusion of the human element in its rationality and explicability.

⁴ Cf. Jacek Dukaj, *Po piśmie* [After writing] (Kraków: Wydawnictwo Literackie, 2019), 132–133. Jacek Dukaj proposes a simple exercise for the imagination: "Imagine you are reading a novel. A traditional novel. Several hundred pages describing the life events of fictional entities, their behaviour, feelings, conversations, decisions. You become immersed in the narrative, identifying yourself with the characters, following the consequences and connections; you live it. The artistic value of this novel is irrelevant; what matters is that it resonates with readers, that you have entered its fictional world to live in it. And then it is revealed that the novel was in fact the product of a program – some neural network trained by Google or Facebook on a vast array of texts representing the entire history of human culture, as well as on trillions of human web searches." Dukaj, *Po piśmie*, 133.

It appears that the multifaceted activity of AI is becoming a part of our daily lives. As a result, the conventional distinctions between “human” and “non-human,” “real” and “fake,” “spirit” and “machine” are gradually losing their significance. The identity of the author or the source of an artwork or text will cease to be of any importance to us. We will not be concerned with whether the content we are interested in originates from the human or non-human *logos*. This scenario is becoming increasingly realistic, as evidenced by the latest generations of programs that are developing on so-called evolutionary algorithms specifically designed to show people what they would like to see.⁵

Considering the examples delineated above, I would hypothesize that the increasing range of possibilities offered by artificial intelligence will in all probability compel schools and universities to undertake urgent reform of their prevailing pedagogical methodologies. I will highlight four challenges that I find most significant, which are likely to be faced by schools and universities in the nearest future. The first concerns the inability to differentiate between authentic and fabricated work produced by students and doctoral candidates. The second is the loss of critical thinking in the age of AI-generated fake-news and hallucinations. The third challenge involves the active assistance of AI in the lecturers’ work, and the last one is the systematic erosion of the significance of compiling the historical achievements into newly created scientific works.

⁵ Cf. Dariusz Prokopowicz, “Rozwój i zastosowania technologii sztucznej inteligencji z uwzględnieniem aspektów etycznych i społecznych” [The development and application of artificial intelligence technologies with ethical and social aspects taken into account], in: *Sztuczna inteligencja. Szanse i zagrożenia* [Artificial Intelligence. Opportunities and Threats], ed. A. Wysocki (Warszawa: Wydawnictwo UKSW, 2024), 104–105; Dukaj, *Po piśmie*, 129, 143; “On constructing and using algorithms in AI” see: Krzysztof Rybiński, Jarosław Królewski, *Algokracja. Jak i dlaczego sztuczna inteligencja zmienia wszystko* [Algocracy. How and why artificial intelligence is changing everything]. (Warsaw: PWN 2023), 31–78.

Algorithms Will Write Our Theses for Us

The first major challenge confronting the academic community is the presence of fabricated academic work produced by students. Nowadays, the employment of artificial intelligence has become pervasive and is widely relieving us of some of our intellectual competencies. With various assignments being delegated to this technology, we are giving up creative thinking, discovering, source investigation, reflection, and the engagement in intellectual effort. Adequate query formulation in the utilization of advanced language models such as those driven by AI LLMs (Large Language Models) or GPT chatbots, is sufficient to obtain, with increasing facility, satisfactory responses, cleverly written essays, translations, research papers, and term papers.⁶

In my opinion, the widespread use of such technological tools will result in a significant number of people relinquishing their creative competence, as they will be delegating a substantial proportion of their cognitive exertion (i.e. reading, comparing, analyzing, and drawing conclusions) to machines. However, it should be noted that in the current state of higher education, in the humanities (e.g. Polish studies, pedagogy, theology, cultural studies, political science, security) students are required to write papers, because the awarding of academic degrees—bachelor’s, master’s, doctorate—is based on these writing assignments. The increasing importance of artificial intelligence, its

⁶ Cf. Zhicheng Lin, “Why and how to embrace AI such as ChatGPT in your academic life,” *Royal Society Open Science*, vol. 10 (2023): 2–11. <https://royalsocietypublishing.org/doi/10.1098/rsos.230658>; Steven Stolz at all, “Is learning with ChatGPT really learning?” *Educational Philosophy and Theory*, no. 12 (2024): 1256–1262: (https://www.researchgate.net/publication/382465858_Is_learning_with_ChatGPT_really_learning) .

growing popularity, and its proficiency in text creation will soon present us with a very serious problem: the inability of distinguishing AI-generated (or mostly AI-generated) texts from those written by a student as the result of collecting materials, laborious research in libraries, reading, correcting, analyzing, and creatively presenting topics understood, information discovered, and conclusions drawn. What used to be a student's or doctoral candidate's independent effort has now become the result of their "cooperation" with AI. Needless to say, the majority of this "work" will be performed by artificial intelligence, including the collection of material, the creation of a bibliography, the arrangement of sentences, the analysis of the content, and even the drawing of conclusions.⁷

The integration of artificial intelligence into educational settings is precipitating a paradigm shift in the nature and demands of academic work. Until now, the composition of substantial academic papers required months, and in case of doctoral dissertation, years of research and drafting. Today, a paid program equipped with extensive databases and advanced processing capabilities, can generate billions of written forms per second, depending only on the processing power put at its disposal. AI tools are unsurpassed in terms of speed of analysis and production. Naturally, the student will improve what they received from the "thinking machine," taking into account the supervisor's comments, but they will ultimately submit the resulting text as their own bachelor's, master's or maybe even doctoral thesis. A work created in such a manner will effortlessly pass plagiarism detection, result-

⁷ It should be observed that the practice of automatic writing of articles was initiated as early as 2014, with Automated Insights publishing approximately one billion articles written by computers as part of the Turing test, the objective of which was to ascertain whether journalists could differentiate between machine-generated and human-generated text. A decade later, it can be concluded that AI has passed the Turing test in this field. Walsh, "It's Alive!," 176.

ing in the bestowal of the academic degree. However, it should be noted that the thesis in question will not be the student's original work, but rather a clever combination of compilation and plagiarism. It is erroneous to assume that an individual or an anti-plagiarism software can accurately identify a fraud, even with the current advancements in AI, which are still very challenging to implement. Machine-generated text is inherently unique, making the software designed for fraud detection essentially ineffective. Consequently, while students may learn little or nothing, they will complete their studies and obtain a degree.⁸

Regulations Against Plagiarism

In such circumstances, educational institutions will face a substantial challenge in establishing regulations and educational standards that can effectively prevent the awarding of degrees to individuals engaging in academic fraud. Concurrently, these institutions should endeavor to create an environment conducive to authentic creative work for students majoring in the humanities.

Universities worldwide have previously attempted to implement such regulatory measures, including my own institution of higher learning, the Cardinal Stefan Wyszyński University in Warsaw. The Rector's Ordinance dated October 25, 2023, stipulates:

⁸ If a computer program can control the rather complicated process of writing articles, effectively coordinate the work of dozens of people, and achieve good results, it means that the machine can efficiently produce various texts on a mass scale. See: Christof Royer, *Outsourcing Humanity? ChatGPT, Critical Thinking, and the Crisis in Higher Education*, "Studies in Philosophy and Education," vol. 43 (2024): 480–484; Bettina Berendt at all, *AI in education: learner choice and fundamental rights Learning*, "Media and Technology", no. 3 (2020): 313–322.]

Content generators can create, for example, essays or research papers that may be mistaken for student work, thus resulting in plagiarism. In addition, there is the issue of the inappropriate use of resources. It should be noted that AI (artificial intelligence) models have limited knowledge, which may be outdated or incomplete. Furthermore, AI models have been observed to hallucinate (fabricate information in such a way that it appears factual to the recipient). In particular, the AI's way of reasoning (response generation style) is perceived as more human-like, however, it may transpire that both the conclusions drawn by AI are false and the sources provided by AI do not really exist (just were generated for answering purposes). Hallucinations pose a significant challenge in scientific domains that demand multi-stage reasoning, as a single incorrect interpretation by AI can substantially impact the veracity of the entire argument. AI tools are devoid of ethical principles and demonstrate a lack of discernment between truth and falsehood. These tools systematically collect and analyze data from databases and web sources, thereby acquiring not only the factual content but also any inaccuracies present in the sources. Consequently, AI can be used to manipulate data, generate unsuitable content, or mislead people. Due to the processing of substantial volumes of data, generative AI models in particular can have ethical and ideological implications concerning, amongst other things, discriminatory content, political views, business practices, ideologies, and so forth. In addition, there are biases arising from the internal codes of ethics and legal regulations of companies providing AI technologies, as well as the private views of the creators of AI algorithms themselves.⁹

⁹ Ordinance No 57/2023 of the Rector of Cardinal Stefan Wyszyński University in Warsaw of October 25, 2023. on the principles for the use of artificial intelligence-based tools in teaching and learning: <https://www.google.pl/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&ved=2ahUKEwjO5JKoq7yDAXVaQvEDHWqEAmcQFn>

The rector recommends that students' work be thoroughly checked and that the preparation of credits in the form of oral presentations or written work in class be encouraged in order to verify learning outcomes and the independence of their work (Ordinance No 57/2023 of the Rector of Cardinal Stefan Wyszyński University).

This problem has also been noticed by rectors and lecturers at other universities. The authorities of one of the most prominent state educational institutions in the Czech Republic, the Prague University of Economics and Business (Vysoká škola ekonomická v Praze), and notably its Faculty of Business Administration, have resolved that their students shall not be required to submit a bachelor's thesis. This resolution was prompted by the uncontrolled activity of artificial intelligence. Presidents and deans of these institutions are aware of the fact that I have already mentioned, namely that online content generators are capable of producing the major part of a student's thesis and that reviewers are often unable to detect plagiarism. Consequently, it has been decided to implement a new solution, whereby students are required to do an internship, participate in a faculty research project, or complete a specific business program as a component of their bachelor's degree. Another unit that is modifying its existing curriculum as a result of the emergence of AI text generators is the Faculty of Law at the University of Brno, where classic written assignments are being replaced with presentations and projects that require the direct involvement of students in the creative process.¹⁰

oECBYQAQ&url=https%3A%2F%2Fmonitor.uksw.edu.pl%2Fdocs%2Fdownload%2F9f83a8da2c44b9816f594d502868b542&usg=AOvVaw1wYqDmJzxtTVxAzuJLPibs&opi=89978449 (accessed: March 2025).

¹⁰ See: *Rules and guidelines for using generative artificial intelligence tools*: <https://fm.vse.cz/sluzby-pro-studenty/pravidla-pokyny-genai-studenti/>; *AAIN Generative Artificial Intelligence Guidelines in Australian Academic Integrity Network (AAIN)*: <https://www.teqsa.gov.au/sites/default/files/2023-04/aain-generative-ai-guidelines.pdf> (accessed: March 2025).

Critical Reasoning vs. Algorithmic Hallucinations

These examples illustrate that the younger generations are in danger of losing their intellectual and conceptual competence due to widespread fraudulent activities. This process has serious consequences, potentially leading to a loss of intellectual capacity and a consequent inability to perceive reality accurately and critically, given the close relationship between the two. The ability to think and create is inextricably linked to the capacity to accurately and carefully perceive the content presented by the world, both human and digital. The accelerated development of algorithms that emulate human activity, along with the emergence of devices capable of generating texts, music, and images, may potentially compromise the ability of young individuals to reason in a manner that is resistant to falsehood, particularly in the context of digital AI modes.

To prevent this, it is imperative to acknowledge that educational institutions, particularly secondary schools and academic centers, should focus on reinforcing their efforts to teach students independent analytical thinking skills based on source verification and argumentation. These competencies will become increasingly important in the context of AI-generated content, which already permeates various digital and informational domains, including the web, news outlets, public spaces, and journalistic discourse. The capacity of artificial intelligence to replicate human-like voices and produce deceptively lifelike images and videos makes it an effective tool in creating fake information.¹¹ The Ordinance of Rector of the UKSW, as quoted above, men-

¹¹ Cf. Stanisław Jędrzejewski, Krzysztof Kuźmicz, “Informacyjna (nie)rzeczywistość: wybrane myśli i obserwacje” [Informational (un)reality: Selected thoughts and observations], in: *Przyszłość jest dziś. Trendy kształtujące biznes, społeczeństwo*

tions yet another phenomenon, designating it succinctly as “hallucinations.”¹²

What does this word mean in the context of AI behavior analysis, i.e. can machines hallucinate? The term has already entered the dictionary of computer science and means an error in algorithms arising from a computer’s inability to discern the right reasons, to distinguish material truth from falsehood, and from its ability to create non-existent data.

These mistakes occur because machines are incapable of weighing reasons and interpreting data correctly. They can only perform mathematical operations on the data that they have been fed. Their functioning is based on a logical, and therefore non-contradictory, albeit not necessarily truthful (in the sense of factuality) perception of a given content (e.g., a three-headed bird or a two-tailed reptile). A machine that generates hallucinations meticulously organizes content based on truthful data obtained from a programmer, the Internet, and databases. However, it also arranges the content that is coherent but not necessarily true, such as footnotes from non-existent books, authors, or publications, therefore creating non-contradictory but inaccurate information.¹³

i przywództwo [The future is now. Trends shaping business, society, and leadership], ed. M. Ciszewska-Mlinaric (Warsaw: Akademia Leona Koźmińskiego, 2023), 301–302. One of many examples of the real possibilities of AI is the creation of increasingly better videos impersonating public figures (actors, politicians, clergymen), as well as photos, often of a compromising nature. See: <https://reutersinstitute.politics.ox.ac.uk/news/how-ai-generated-disinformation-might-impact-years-elections-and-how-journalists-should-report> (accessed: March 2025).

¹² Ordinance No 57/2023 of the Rector of the UKSW.

¹³ Cf. Yoni Wilkenfeld, *Can Chess Survive Artificial Intelligence?* “New Atlantis” no 58 (2019): 37–45; Jacek Grzybowski, “Czy maszyny myślą? O definiowaniu intelektu i inteligencji w dobie rewolucji cyfrowej” [Do machines think? On defining intellect and intelligence in the age of the digital revolution], *Thomistic annual* no. 13 (2024): 114–117; Leszek Korporowicz, “Sztucznej mądrości nie będzie” [There will be no artificial wisdom], *Perspectives of culture* no. 3 (2023): 124.

If rationality, as Stanisław Judycki states, consists in performing operations on reasons, then machines cannot do that, since reasons are not physical beings defined by their properties, such as momentum and mass. Reasons are immaterial, and as such, the entity that utilizes, organizes, examines, compares, and reflects on reasons must be a conscious entity. Therefore, the judgment of reasons does not constitute a physical operation; rather, it is a subjective process that, to the best of our knowledge, is exclusively executed by the human intellect.¹⁴ Within the cognitive process of the human individual, it is possible to recognize the existence of something (existential judgment) and subsequently affirm the veracity, or conformity with reality, of the content of this really existing object (veracity analysis). This capacity, at least to the present, remains beyond the capabilities of any computer program.¹⁵ It operates on data, arranging non-contradictory sequences of meaning; however, it occasionally arranges content in an apparently logical but ultimately false manner – a phenomenon often referred to as “hallucination.”¹⁶ If a machine was able to decide on the reasons and goals of its actions and to evaluate them, it would become autotelic, which means its purposes would be self-contained. However, there is currently no evidence of such a capability, which explains the tenden-

¹⁴ Cf. Stanisław Judycki, *Epistemologia* [Epistemology], vol. 1 (Poznań–Warszawa: Wydawnictwo W drodze, 2020), 97–98.

¹⁵ Cf. Marek Dobrzeński, *Deficyt zaufania do świata. Polska filozofia wobec współczesności* [Deficit of trust in the world. Polish philosophy vis-a-vis modernity] (Warsaw, PIW, 2024), 221.

¹⁶ Companies are already emerging in this field (such as Factiveverse in Norway and RagaAI in the USA) that develop tools for verifying content generated by algorithms and AI Chats, with the objective of tracking down and removing hallucinations created by artificial intelligence (<https://factiverse.ai/>; <https://raga.ai/>; Urszula Lesman, *Sztuczna inteligencja lubi kłamać. Bolesne wpadki firm* [Artificial intelligence likes to lie. Painful business blunders]: <https://cyfrowa.rp.pl/technologie/art40297371-sztuczna-inteligencja-lubi-klamac-ufanie-jej-moze-byc-bardzo-kosztowne>).

cy of AI systems to make errors when attempting to complete tasks, generate content, and hallucinate without being aware of it.¹⁷

I believe that the way to avoid succumbing to such trends is to check sources, verify information, and develop the ability to reason critically and independently. Hence the idea, still relevant and advocated by many philosophical circles, that philosophy should be a mandatory component of secondary education.¹⁸ The demonstration of fundamental logical fallacies, the analysis of provided content, and the capacity to argue and think critically are competencies that appear to be indispensable in an era when we are increasingly subjected to manipulation, falsehood, disinformation, and, naturally, AI hallucinations, precisely due to the potential of AI. Academic papers are already drawing attention to this: “Excessive reliance by students on AI tools can lead to a lack of development of problem-solving skills and critical and creative thinking.”¹⁹

¹⁷ Ray Kurzweil, *How to create a mind. The secret of human thought revealed* (New York: Viking Penguin, 2012); Nick Bostrom, *Superintelligence: paths, dangers, strategies* (Oxford: Oxford University Press, 2014).

¹⁸ For example, in the Second Polish Republic, a subject called “propaedeutics of philosophy” was mandatory in the secondary school curriculum. In 1948, the communists deliberately abolished these lessons during the process of Stalinization of the Polish education system (Jakub Berman’s decree). They understood that teaching critical thinking and freedom of thought was a threat to their ideology and politics. Following the restoration of Poland’s independence in 1989, philosophy as a component of the curriculum was not reinstated in schools. Why was no attempt made to reintroduce the subject as a mandatory component of secondary education (as was the case in France, for example)? It is difficult to answer this question today. Ultimately, the absence of philosophy in the general education curriculum is a form of discrimination, as it is accorded a significantly lower status compared to other subjects. This disparity is particularly noticeable when considering the manner in which philosophy is perceived by decision-makers, who often regard it as a counteragent to dogmatism and ideology. Cf. Jacek Grzybowski and all, *Sposób na filozofię. Kluczowe zagadnienia z dydaktyki przedmiotowej* [The way to philosophy. Key issues in subject didactics] (Warsaw: Liber Libri, 2016), 41–42.

¹⁹ Ordinance No 57/2023 of the Rector of the UKSW.

Artificial Intelligence in Research Activities

The third problem is the way in which AI is used in university work. Anyone who follows trends in higher education has observed that for some time now, students, doctoral students and university lecturers have been regular recipients of numerous emails offering training in the use of artificial intelligence and GPT Chat. These emails cover a wide range of topics such as: *How to use AI and GPT Chat to write scientific papers?*; *Use of AI in scientific activities – legal and ethical aspects*; *AI in the researcher’s activities – tool overview (writing papers, resources, projects, presentations, data analysis)*. Training courses that focus on utilizing the potential of AI are being offered, and all of these programs have one clear message in common:

The ability to use artificial intelligence is becoming one of the key elements of success, not only in business but also in the field of science. Those who master the language of algorithms more promptly, will enhance their efficiency and save time and energy.²⁰

To substantiate the necessity of acquiring the aforementioned competencies, course developers (often with the academic titles of doctor or professor) formulate the reasons for using artificial intelligence as follows:

²⁰ *Sztuczna inteligencja (AI) w działalności badacza – przegląd narzędzi (pisanie publikacji i zasoby, projekty, prezentacje, analiza danych* [Artificial intelligence in research activities—review of tools (writing publications and resources, projects, presentations, data analysis)]: <https://centrumidea.pl/szkolenia/szkolenia-otwarte/nauka-i-szkolnictwo-wyzsze/sztuczna-inteligencja-ai-w-dzialalnosci-badacza> (accessed: March 2025).]

In this practical and interactive training course, you will understand how AI works, what it can do, and what it cannot (yet) do. You will find out how to accelerate research and data analysis, and you will learn techniques for creating publication plans and writing texts, paragraphs, abstracts, and summaries in collaboration with the GPT Chat and other tools. You will learn how to plan and write articles and other academic publications using AI and you will discover how to make better use of AI for writing texts, abstracts and conclusions.²¹

The instructors aim to show that generative artificial intelligence is a key element in the set of tools used by researchers. These tools can be used by researchers in all fields and disciplines, for example, to work with text (e.g., writing papers or grant proposals), but also to process data, generate visualizations, create presentations and slides, and manage teams. The above-mentioned paid training courses, of which there are more and more on the education market, try to show researchers how much easier they can make their work and how much time they can save.

The abundance of such offers means, in my opinion, that artificial intelligence will quickly become an important (or perhaps shortly indispensable) tool in scientific work. The above-mentioned training courses are designed to make scientists aware that they should increase their scientific efficiency by mastering AI tools. Therefore, it is imper-

²¹ *Pisanie artykułów (publikacji) naukowych: narzędzia AI, unikanie nieetycznych i zabronionych praktyk generatywnej sztucznej inteligencji, specjalistyczne edytory, współpraca z wieloma autorami* [Writing scientific articles (publications): AI tools, avoiding unethical and prohibited generative AI practices, specialist editors, working with multiple authors]: <https://centrumidea.pl/szkolenia/szkolenia-otwarte/nauka-i-szkolnictwo-wyzsze/pisanie-artykulow-naukowych-wieloautorskie> (accessed: March 2025).

ative to learn the “chat” language and techniques for formulating *prompts* (sentences and questions that are instructions or commands to the AI system to help it analyze a specific set of data) and obtaining contextual *responses* (answers that are truthful and correspond to reality). These skills enable us to effectively “talk” to thinking machines. It also helps to learn how to use GPT Chat for *brainstorming*, so that you can actively generate solutions and ideas for constructive and contextual queries. There is no way around it, the trainers argue, you simply have to learn to work with AI to quickly and efficiently obtain the information and data you require. The ultimate objective of this training is to persuade us of the efficacy of artificial intelligence tools in every stage of the scientific publication process, including the generation of ideas on a chosen topic, assistance in planning and writing articles, and even the analysis of the influence exerted by scientific activities, including the description of the impact of published research results or research and development activities. The instructors demonstrate to us the potential of AI to function as a comprehensive repository of specialized knowledge, combined with the best innovative activities and practical advice in a given domain. Interestingly, the proposed courses also include an ethical dimension. Their objective is to show the negative aspect of using artificial intelligence such as the generation of substandard “scientific products” or even plagiarism.²²

The active incorporation of the possibilities offered by AI into scholarly work therefore seems to be a sign of the times. It remains to be seen though, whether educators and scholars will resist the allure of compiling and relinquishing the burden of creative work, now that dig-

²² See: *Wykorzystanie sztucznej inteligencji w nauczaniu akademickim: od planowania zajęć po indywidualizację procesu dydaktycznego* [The use of artificial intelligence in academic teaching: from lesson planning to individualization of the teaching process]: <https://centrumidea.pl/szkolenia/szkolenia-otwarte/nauka-i-szkolnictwo-wyzsze/ai-w-nauczaniu-akademickim> (accessed: March 2025).

ital technology has made it possible. Will they reproduce the mistakes of their students by delegating part of their creativity to AI?

The Machine Will Tell a Story

The fourth challenge is associated with the collection of historical data and the presentation of accomplishments in a given field to date. In technical terms, this is referred to as the state of research. I believe that in the age of digitalization, a substantial and time-consuming part of scientific work (the collection of materials, the exploration of sources, and the presentation of what has been done, discovered and understood up to date) is losing its meaning due to the rapid advancement in artificial intelligence, which made possible the expeditious compilation and assembly of data. Today, a machine will provide the overview of historical research, create a bibliography, summarize historical accounts, and even formulate preliminary conclusions.

What are the implications of this shift for academic writers? It should be acknowledged that the sections of our books that have been crucial to date as they demonstrated our erudition, knowledge of history, and the state of research, will no longer be essential. These elements will be increasingly performed by artificial intelligence. While this form of learning, cognition, and understanding will remain essential for scientists, it will no longer be a component of independent scientific research and will no longer be duly appreciated. Historical information on various approaches to a particular problem, such as the problem of truth from the Presocratics to the postmodern era, will be gathered by AI from sources, bibliographies, and annotations. Consequently, a significant portion of research in the humanities, which serves as a sort of historical prelude to formulating and initiating one's own solutions, will no longer be required in the form of text. While it will remain relevant for personal research, it will lack the aca-

demic weight it has had for centuries. Analyzing this AI competence, I believe that, paradoxically, it is precisely this sphere of its development that will finally give meaning to our own interpretations, ideas, and concepts that we will propose after acquiring knowledge and understanding of the previous solutions, but no longer including them in our work. What will gain significance is the application of human intelligence, brilliance, and ingenuity, a creative approach to problem-solving that is inventive and new, that cannot be found in history and will certainly not be “invented” by machines.

“When Thought Is Fascinating For Its Own Sake”

To summarize, let us pose the question: how should the academic community, especially the humanities, respond to the outlined issues? This includes the problem of producing fabricated papers, the dangers of disinformation and manipulation, the controversies surrounding the creative work of AI, and, finally, the diminishing importance of historical summaries. In view of the imminent technological and social changes, what educational program should we pursue?

It is my belief that the methods and ideas implemented at medieval universities, and especially the practice of the liberal arts (*septem artes liberales*), can be of help in our current situation. These methods originate from an ancient educational curriculum based on comprehensive teaching divided into two phases. The first phase emphasized grammar, dialectics, and rhetoric, while the second phase was focused on the exact sciences, including arithmetic, geometry, astronomy, and music.²³ The

²³ The foundations of the liberal arts were established in the 5th century AD by Marcus Apuleius, an African pagan writer, in his handbook *Satyricon, sive de nuptiis Philologiae et Mercurii et de septem artibus liberalibus*. In the early Middle Ages his

educational program established in this way (the *trivium* and *quadrivium*) taught young people to ask questions, to discuss, to formulate theses and conclusions, and to critically examine the views of others.

The pursuit of such ideals gave rise to medieval quodlibets—free discussions and debates held regularly at universities. These events attracted large audiences, which was facilitated by the prevailing form of teaching at the academies of the time: the disputation (*disputatio*) was a type of exercise conducted before the main lecture, wherein the subject matter of the lecture was deliberated. Medieval scholars often engaged in disputes during lectures, yet twice annually they also organized the so-called *disputationes quodlibetales*—solemn university debates. Distinguished guests were invited to present a paper, while lecturers and students from the entire academic community came to participate in the debate. After the lecture there was time for asking questions on various subjects as well as discussing various problems. It is noteworthy that during the 12th and 13th centuries, medieval universities were a place of unparalleled freedom in discourse—a quality that contemporary universities would do well to emulate.²⁴

work, edited by Cassiodorus (485–583), was widely available and read in nearly every library throughout Europe. This resulted in the seven liberal arts—the trivium (basic subjects: grammar, rhetoric and logic) and the quadrivium (advanced subjects: arithmetic, geometry, music and astronomy)—becoming the predominant subjects of study in the medieval period. These educational models placed particular emphasis on the canon of ancient literature, classical languages (Greek and Latin) and the comprehensive memorization of numerous literary, poetic, philosophical and theological works. The most illustrious geniuses, writers, humanists, explorers and notable persons of many centuries were educated in the tradition of the European canon of teaching forged in this manner. See more: Werner Jaeger, *Paideia: The Ideals of Greek Culture*, vol. 2, *In search of the divine centre*, tr. Gilbert Highet (Oxford: Oxford University Press, 1986), 237–244; William P. Ker, *Medieval English Literature Home University of Modern Knowledge*, tr. Willard R. Trask (London: Oxford University Press, 1969), 92–95; Ernest R. Curtius, *European Literature and the Latin Middle Ages* (Princeton: Princeton University Press, 1990), 56–58.

²⁴ See more: Leonard E. Boyle, “The Quodlibets of St. Thomas and Pastoral Care,” *The Thomist* no. 2 (1974): 232–56; John W. Wippel, “Quodlibetal Questions, Chiefly in

University education is supposed to transform students into a new kind of person, who can think, enquire, critically analyze the world and ideas, formulate judgments, apply sound reasoning, compare and reflect on the contents of various branches, areas, and fields of academia. This is the true mission of the university and solid knowledge derives from it.²⁵ John Henry Newman argued that the proper task of a university is to employ itself in the education of the intellect, making this intellectual culture its direct scope. It educates the intellect to reason well in all matters, to reach out towards truth, and to grasp it. Within this model, a university community is something more than a mere gathering of pragmatists—it is a *koinonia*, a community of values and goals, in which truth and goodness constitute the binding power of harmony, the supreme ideas being the paragons that motivate human cognitive pursuits.²⁶ Therefore, the special social role of the university should be emphasized, since it is to equip its students with the most comprehensive knowledge and a general humanistic education, laying the foundation for cultural and moral strength. The university should provide society with enlightened and independent thinkers. We learn to create because only this ability makes us truly human. Without creativity, we are merely a “pathetically, if cleverly, arranged machine.”²⁷

Theology Faculties,” in: B.C. Bazàn, G. Fransen, J.W. Wippel, D. Jacquart, *Les questions disputées et les questions quodlibétiques dans les facultés de théologie, de droit et de médecine* (Tournhout: Brepols, 1985), 151–222; *Theological Quodlibeta in the Middle Ages. The Thirteenth Century*, ed. Ch. Schabel (Leiden–Boston: Brill, 2006); Marie-Dominique Chenu, *Aquinas and his role in theology*, tr. Paul Philibert (Collegeville, Liturgical Press, 2002), 78–80.

²⁵ Cf. Alasdair C. MacIntyre, *God, philosophy, universities: Selective history of the Catholic philosophical tradition* (Lanham, Sheed and Ward Book/Rowman & Littlefield Publishers, 2009), 23–25.

²⁶ Cf. John H. Newman, *The Idea of a University* (Washington, Regnery Pub., 1999), 203.

²⁷ Clemens Menze, *Wilhelm von Humboldt*, tr. T.G. Pszczółkowski (Warszawa: Wydawnictwo DiG, 1993), 7–9.

Also in the modern era, the German educational reformer Wilhelm von Humboldt (1767–1835) looked to ancient culture for the sources of such a model of education. He argued that studying the ethos of the ancients would benefit students in many ways:

In addition to individual studies and education, there is another very distinctive study that concerns the whole person. This kind of education is becoming increasingly rare, but it was once very common in ancient Greece. I don't think it can be revived any better than by studying great and formidable people, and by that I mean studying the Greeks.²⁸

Philosophy, history, and art are of particular importance in this educational model. Philosophy enables individuals to transcend their subjective impressions and learn to think abstractly and theoretically, history enables them to understand the reasons for the successes and failures of previous generations, and art gives them a sense of creativity, development, and self-expression.

Where thought fascinates for its own sake, there a truly scientific sense directs thinking to its original sources. Where thought is used for purposes beyond itself, there science may exist, but its spirit dies. The various forms of manifestation of scientific needs, reduced to their simplest essence, invariably incorporate the recognition of the hidden in the visible.²⁹

²⁸ Bolesław Andrzejewski, *Wilhelm von Humboldt* (Warszawa: Wiedza Powszechna, 1989), 94–113 (trans. Jacek Grzybowski).

²⁹ Wilhelm von Humboldt, *O myśli i mowie. Wybór pism z teorii poznania, filozofii dziejów i filozofii języka* [On thought and speech. Selected writings in epistemology, philosophy of history and philosophy of language], ed. and tr. E.M. Kowalska (Warszawa: PWN, 2002), 6.

Such a university, drawing from ancient and medieval sources, facilitates the development of models of educational work that assist in monitoring students' intellectual growth, creativity, understanding, and acumen. It does not permit mere copying. It teaches reading texts, discussion, argumentation, and critical reasoning. In classes conducted in this manner, learning is not intended for short-term, non-scientific purposes; rather, it is pursued out of a profound fascination with the very idea of learning itself. This pedagogical approach fosters a sense of curiosity about the world and personal development.³⁰

University of the Future

In response to the challenges posed by technology to the field of education, which I have previously outlined, and others that have not yet been described, the fundamental concern of our universities lies in the education of individuals who are well-read, capable of thinking, unafraid to ask the most unusual questions and to propose unconventional solutions. If universities are to effectively prepare intellectuals to serve society, the issues raised here require a university that places strong emphasis on philosophy, a broad spectrum of ethics, the history of anthropological theories, philosophy of culture and philosophy of politics (social and cultural theories along with their consequences) and a significant number of workshops where source texts are discussed, debated, arguments are formulated and refuted. Finally, under the guidance of a supervisor, students prepare an essay or a presentation on a specific topic, thus learning how to present their thoughts in a well-structured manner.³¹

³⁰ Humboldt, *O myśli i mowie*, XXIX.

³¹ Tadeusz Czeżowski, "Dydaktyka filozofii w nauczaniu uniwersyteckim" [The didactics of philosophy in university teaching], in: Czeżowski, *Odczyty filozoficzne* [Philosophical lectures], (Toruń: PWN, 1958), 311–312.

In this manner, it is possible to create universities that are rich in culture and free of ideology, because a solid foundation of knowledge together with an emphasis on discussion, research, communication, independent argumentation, learning methods of objectification and the use of a meta-language, will support the formation of people who think freely and carry out scientific projects that are not the product of technology.

Why should we approach didactics in this way? For at least two reasons. In the first place, we can observe the students' activity and their scientific development so as to avoid academic fraud and the awarding of academic degrees to those who do not deserve them. After all, the people who finance university education do so to cultivate individuals who will not only abstain from falsifying their work by using the operational power of AI to quickly join the ranks of business executives, but will also be prepared for responsible social service.³² Secondly, the

³² There is no exaggeration in saying that modern universities are increasingly regarded as public sector organizations rather than traditional and historically exceptional institutions. They are subjected to the intense pressure of evaluation which confronts their efforts with a permanent cost-effectiveness ratio calculation and enforces competitive activities in the field of public financial resources. For this reason, most educational pathways seek to establish a connection between a university degree and labor market trends. In the Anglo-Saxon world, the concept of the "entrepreneurial university" has emerged. These trends were already identified at the beginning of the 20th century by Max Weber—his essay *Science as a Vocation* revealed that universities began to resemble capitalist (that is organized, bureaucratic, goal-oriented) enterprises. As a result of rationalization processes, the desacralization of culture, and the "disenchantment of the world," the concept of "vocation" was gradually abandoned in science. See: Max Weber, *Essays in Sociology*, tr. and ed. H.H. Gerth, C. Wright Mills (New York: Oxford University Press, 1946), 130–132. From the perspective of liberal thought and neoclassical economics, education becomes an effective factor of economic growth when it provides skills that are in demand on the market. Education should therefore be tailored to labor market needs, both from the standpoint of employers as well as students and their parents. Cf. Burton R. Clark, *Creating Entrepreneurial Universities. Organizational Pathways of Transformation* (New York: IAU Press, 1998).

most serious debates and pressing socio-political challenges facing our societies include the rapid development of digital technology, widespread job losses, the erosion of democratic culture, electoral interference, growing control exercised by society, digital exclusion, and ethical dilemmas posed by biotechnology and genetic engineering methods that allow substantial modifications of the human genome.³³ Undoubtedly, the problem of migration will also intensify, along with the risk of legitimizing armed conflicts. Therefore, we need a university model where the teaching system is not based on entrepreneurship, but on an erudite and creative problem solving. If, during their university studies, students were to have the opportunity to spend several years in a free atmosphere, under the guidance of well-educated staff, engaged in reading classic texts from various fields, debating, arguing, and developing their argumentative reasoning, it could equip them with the skills needed to help discuss and solve pressing dilemmas that will soon arise in our communities. Without humanities graduates, there can be no valid, intelligent and responsible discussion of issues that are typically ethical and social rather than technical. These are problems our world will be facing in the nearest future and, undoubtedly, the subject of debate in decades to come.

A true university is a “university of culture,” where classics and their way of thinking can be studied in the lecture hall and where scholars and students can engage in a free debate on important issues in the practicum. This model of education, although it may sound idealistic, can implement models of humanity that invigorate culture and

³³ Contemporary inquiries into the implications of modern biotechnology, its potential ramifications, and, most notably, its ethical acceptability are primarily initiated by philosophers such as Jürgen Habermas (*The Future of Human Nature: Are We Heading Toward a Liberal Eugenics?*), Michael Sandel (*The Case Against Perfection: Ethics in the Age of Genetic Engineering*), and Martha Nussbaum (*Human Dignity and Bioethics*).

socially emancipate ever wider circles of intellectually gifted people into the role of conscious citizens. Of course, I am aware that this requires the development of a new scientific culture, but if universities do not begin to develop such a culture, they will either become mere suppliers of qualified personnel for the technology business or lose their social prestige altogether. The focus should therefore be on the humanities in the broadest sense, rather than computer science and law, which have unfortunately become the key disciplines of our time.³⁴

To Teach Reasoning Skills and Competencies

A critical examination of the potential contributions of artificial intelligence (AI) to the realm of education offers a valid starting point for a discourse on the current state of our educational system and the responsibility entrusted to those tasked with the design of curricula for

³⁴ Many observers note that those individuals who uncritically trust in the development of modern technology support a narrative that increasingly undermines factual and critical reflection on the development of technology. Techno-enthusiasm clearly has features of ideology, and today's techno-enthusiasts are often people who only see the future in a positive light, without calculating what technology takes away from us and how it changes us for the worse. Today, we are increasingly dealing with networked technology that constantly calls us back (absorbs our attention and time) and is highly addictive. We do not yet know how this will affect the lives of future generations. See: *Cyber kontra real. Cywilizacja w techno-pułapce. Z Andrzejem Zybertowiczem rozmawia Jarema Piekutowski* [Cyber vs. real: Civilization in the techno-trap. A conversation between Andrzej Zybertowicz and Jarema Piekutowski] (Warszawa: Fundacja Nowa Rzeczpospolita, 2022), 12–19; Max Fisher, *The Chaos Machine: The Inside Story of How Social Media Rewired Our Minds and Our World* (London: Quercus, 2023), 48–65; Sylwia Jaskuła, “Sztuczny nauczyciel – prawdziwe konsekwencje wychowawcze” [The artificial teacher—Real educational consequences], in: *Sztuczna inteligencja. Szanse i zagrożenia* [Artificial Intelligence: Opportunities and threats], ed. A. Wysocki, (Warszawa, Wydawnictwo UKSW, 2024), 138–141.

the incoming generations of students. In light of the exponentially increasing conceptual power of AI and its current capabilities, which include generalized information processing, deduction, induction, and the perception of logical connections between propositions, it is imperative to recognize that, employing the definition proposed by Kazimierz Ajdukiewicz and Stanisław Judycki, we are confronted with a distinct manner of cognizing the world through semantic content.³⁵ Consequently, within the humanities, a different approach to teaching methodologies and a new assessment system are imperative. If our objective is to genuinely educate people and cultivate their cultural competencies, we cannot keep establishing business schools that are contingent upon the economic and political landscape, prioritizing pragmatism and the pursuit of profit. Instead, we should promote universities that are rich in content and are able to teach critical and creative reasoning skills. While technology undoubtedly offers numerous benefits, it concomitantly deprives us of something. For every advantage it brings, a corresponding disadvantage can be identified, and the more significant the achievements of technology, the more substantial the negative consequences for people and society.³⁶

³⁵ Operations performed by machines that imitate human reasoning are induction (a type of reductive reasoning that involves inferring the truth of a given universal proposition from the truth of its particular premises—drawing general conclusions from specific cases), deduction (reasoning in which a strict conclusion is drawn from a previously assumed set of premises, entirely contained within its assumptions and not requiring the creation of new statements or concepts—a simple drawing of conclusions), inference (moving from previously established statements/propositions to a new, not acknowledged before statement/proposition—a conclusion). See: Kazimierz Ajdukiewicz, “Język i poznanie” [Language and cognition], vol. 2: *Wybór pism z lat 1945–1963* [Selected writings from 1945–1963] (Warszawa: PWN, 1965), 206–225; Judycki, “Epistemologia” [“Epistemology”] 70–73; David J. Gunkel, *Person thing robot. A moral and legal ontology for the 21st century and beyond*, Massachusetts (Cambridge, Massachusetts: The MIT Press, 2023), 43–44.

³⁶ Neil Postman, *Technopoly: The Surrender of Culture to Technology* (New York: Alfred A. Knopf, 1992), 16–20.

Of course, I am well aware that artificial intelligence is a tool that can help in learning, but it will never fully replace human imagination, sensitivity, and creativity. We should utilize its potential as a tool and teach our students to do likewise, by demonstrating how AI contributes to the development and expansion of our scientific capabilities. Simultaneously, however, we must raise awareness about the perils of delegating our work and efforts to machines. The point was aptly summed up by Pope Francis:

Algorithms must not be allowed to determine how we understand human rights, forgetting the value of compassion, mercy and forgiveness.³⁷

Therefore, it may be necessary to develop a kind of “technological apprenticeship” for pupils and students, as well as to initiate systematic training for teachers and lecturers, helping them to use this versatile tool judiciously, without being exploited by it.³⁸ The objective of these activities would be to do more than just teach how to observe so-called digital hygiene and how to work with new technologies; the objective would be to stimulate broad reflection on the limits of technical progress. It is my conviction that should we neglect to seriously consider this issue, the AI explosion will lead to the implosion of the educational institution. It is imperative to acknowledge that in the era of AI development, original, unique approaches based on human intelligence, brilliance and genius are going to become increasingly impor-

³⁷ Pope Francis, *Artificial Intelligence and Peace. Message of His Holiness Pope Francis for the 57th world day of peace*, 1 January 2024: <https://www.vatican.va/content/francesco/en/messages/peace/documents/20231208-messaggio-57giornatamondialepace2024.html> (accessed: March 2025).

³⁸ Lidia Bielini, *Uczenie się w przestrzeniach cyfrowych uniwersytetu* [Learning in the university’s digital spaces] (Kraków: Impuls, 2022), 181–230.

tant. As I have already pointed out, it is not sufficient to merely study what others have done; it is crucial to make our own contributions, to do something innovative and original. The high suggestiveness and excellent technical quality of machine-generated content poses a challenge, so students should also be taught to critically evaluate this content. Identifying deepfakes will undoubtedly become our most pressing task in the imminent future. We all need to cultivate cognitive caution, to look critically, to be humble, and to meticulously verify the reports we receive. It should be recognized that an accurate understanding of information and its translation into practical applications will become more important than the transmission of information itself.³⁹

Conclusion

We are all well-aware that changes in education must be aligned with changes in the labor market. In the nearest future, jobs that require specialized skills and critical thinking will be preferred and valued, while

³⁹ Robert DiYanni, *Critical and Creative Thinking: A Brief Guide for Teachers* (Oxford: John Wiley & Sons, 2016), 4–27; Ashok K. Panigrahi, Vijay Joshi, “Use of Artificial Intelligence in Education,” *The Management Accountant* no 5 (2020): 64–67. As Pope Francis put it: “Artificial intelligence systems can help to overcome ignorance and facilitate the exchange of information between different peoples and generations. For example, they can render accessible and understandable an enormous patrimony of written knowledge from past ages or enable communication between individuals who do not share a common language. Yet, at the same time, they can be a source of cognitive pollution, a distortion of reality by partially or completely false narratives, believed and broadcast as if they were true.” Pope Francis, *Artificial Intelligence and the Wisdom of the Heart: Towards a Fully Human Communication. Message of His Holiness Pope Francis for the 58th world day of social communications*: https://www.vatican.va/content/francesco/en/messages/communications/documents/2024_0124-messaggio-comunicazioni-sociali.html (accessed: March 2025).

simple manual jobs will be increasingly automated. Numerous studies show the imminence of the labor market transformation under the influence of AI development—it will be necessary to cooperate with machines, constantly educate and expand one's skills as well as adapt to changing technologies. Human work will become less mechanized (routine) and more inventive and original, based on thinking and creativity.⁴⁰ Universities should prepare young people for the current trends in the labor market. Pedagogical communities, aware of forthcoming challenges, are putting forward postulates related to a new vision of education. What will be of primary importance is the dialogical discourse of Western logic and tradition with the cognitive practices of other cultures and traditions, the shift from a literate to a post-literate and post-symbolic culture, and an awareness of the various narratives that permeate our social reality. The university should develop the skills of prospective, interdependent and relational thinking that takes into account the broad context of our lives. This must be followed by the development of a sense of responsibility and agency, the practice of co-presence, team building, dialog, and human-machine collaboration. The enormous expansion of technology must therefore be accompanied by adequate training in responsibility for its development.⁴¹

⁴⁰ Izabela Grabowska, Agata Jastrzębowska, “Megatrendy na globalnym rynku pracy: demografia, kapitał ludzki i migracje” [Megatrends in the global labor market: Demographics, human capital, and migration], in: *Przyszłość jest dziś. Trendy kształtujące biznes, społeczeństwo i przywództwo* [The Future is today. Trends shaping business, society, and leadership], ed. M. Ciszewska-Mlinaric (Warszawa: Akademia Leona Koźmińskiego, 2023): 220–222. Tables and lists of occupations are already being published today, illustrating the elimination of horses and humans from work in the 19th and 20th centuries, while showing which occupations will disappear due to artificial intelligence. Cf. Yuval Noah Harari, *Nexus. A Brief History of Information Networks from the Stone Age to AI* (Great Britain: Fern Press, 2024), 368–385.

⁴¹ In contemporary pedagogical proposals, we find postulates of a departure from the traditional model and system of education. The idea is not to go through successive

Artificial intelligence will undoubtedly grow in importance, but the challenges it poses are not so much technical as anthropological, educational, social and political. AI promises labor savings, more efficient production, more efficient transportation, and more dynamic markets, but at the same time it is revolutionizing the way we live and perceive the world. For this reason, those responsible for education must be particularly aware of the changes underway and how they can be managed with the objective of protecting culture, history, and the uniqueness of human creativity. Artificial intelligence should serve the best of human potential and our noblest aspirations, not compete with them. That is why I believe that education in the age of AI should first and foremost promote critical thinking. Young people, in particular, should develop the ability to use the data and content they find online consciously and critically, while schools, universities and scientific societies must learn to help students and professionals consider the social and ethical aspects of the development and use of technology. Ultimately, it is a matter of reflecting on the role that technology plays in our lives and answering the question of whether human cooperation with AI can

levels/stages of education that end with exams and diplomas, but to focus on the acquisition of specific competencies. In this model, the teacher or lecturer is more of a partner and guide rather than the sole provider of knowledge and then examiner. They assist in navigating the maze of knowledge and information, while teaching creative, critical and free thinking, helping to negotiate the meanders of reality, rather than directly teaching specific information. This model is often implemented in homeschooling and in the so-called “cloud school” project. This means that the traditional step-by-step model of education and acquisition of skills needed later in life may become a thing of the past, and a new model is emerging—continuous independent learning and improvement of skills, frequent changes of occupation and qualifications, readiness for change, openness and flexibility. See: P. Voelkel, *Zawód przyszłości to „bycie pożytecznym” i gotowość do zmian* [The profession of the future is “being useful” and being ready for change]: <https://poznan.wyborcza.pl/poznan/7,36001,30636190,piotr-voelkel-zawod-przyszlosci-to-bycie-pozytecznym-i-gotowosc.html> (accessed: March 2025).

contribute to building a more just and humane world. Universities have their own unique role to play here.



Artificial Intelligence (AI) and the University. How AI Will Change Our Approach to Education

SUMMARY

The extremely dynamic development of artificial intelligence (AI) raises the unavoidable question about the future and direction of education, especially in higher education. The current expansion of AI is leading to situations where we will not be able to verify whether a given work of literature or a piece of art are the result of human effort (i.e. whether they have come into being as a result of human creativity) or whether they have been generated by computer programming algorithms. We are already aware of the impact that artificial intelligence is having on our everyday lives, so we cannot disregard its influence on educational tasks appertaining to schools and universities. Observers of technological changes may rightly ask whether AI development and its takeover of various human competences will not inevitably disavow all work based on reading, acquiring knowledge, inferring, discovering, writing and producing scientific or literary works? Does the development of AI bring us closer to the controversial but probable statement that at some point the will and efficiency of human agents will become unnecessary or even superfluous? Will it not come to the point where the very source of literature or art will be of no importance to us? We will no longer ask who—a human being or a machine—created a certain work, the only evaluation criterion will be its reception and the impression it makes on the recipient's mind. The above considerations lead me to suggest that widening the range of possibilities for using AI will most likely force schools and universities to reform their current teaching methods. I would like to point out what I consider to be the four most significant challenges that we will be facing in school and university environments in the nearest future. The

first is the problem of failing to distinguish between real and fake master's and doctoral theses, the second is the loss of critical thinking in the era of fake news and hallucinations created by artificial intelligence, the third is the active help AI gives to lecturers, and the fourth is a systematic disappearance of the importance of compiling the historical achievements of predecessors in newly created scientific works. At the end of this paper, I will show the model of a university that would be able to meet the challenges described above.

Keywords: artificial intelligence, university, education, Humboldt, algorithms, reform

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